

Checks for Multivariate Normality (MVN)

A summary of the three graphical/informal checks

Background

Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n \stackrel{iid}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where each observation is a p -dimensional vector. We want to check whether this multivariate normality assumption is reasonable.

Summary Table

Check	Formula / Method	What to look for	Notes
lightgray 1. Univariate QQ plots	Plot each of the p variables individually against the theoretical quantiles of $N(0, 1)$	Points fall approximately on a straight diagonal line for every variable	Necessary but <i>not</i> sufficient. Only checks each margin separately, not the joint structure.
2. 50% rule (known $\boldsymbol{\Sigma}$)	Compute squared Mahalanobis distances using the true covariance: $d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$	Approximately 50% of the d_i^2 values should fall below the median of $\chi^2(p)$	Quick single-point check. Requires $\boldsymbol{\Sigma}$ to be known.
lightgray 3. Chi-squared QQ plot (estimated $\boldsymbol{\Sigma}$)	Compute d_i^2 using the sample covariance matrix $\mathbf{S}$: $d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$, then order them	Plot sorted $d_{(i)}^2$ against the quantiles of $\chi^2(p)$. Points should fall on a straight line.	More practical than check 2 as it estimates $\boldsymbol{\Sigma}$ from data. Outliers appear as high-end deviations.

Key Formulas

Squared Mahalanobis distance (known $\boldsymbol{\Sigma}$):

$$d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}}) \stackrel{\text{approx}}{\sim} \chi^2(p)$$

Squared Mahalanobis distance (estimated $\mathbf{S}$):

$$d_i^2 = (\mathbf{x}_i - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}}) \stackrel{\text{approx}}{\sim} \chi^2(p)$$

where the sample covariance matrix is:

$$\mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$$

Important Caveats

- All three checks are **necessary but not sufficient** conditions for MVN. Data can pass all three and still not be truly multivariate normal.
- Checks 2 and 3 are closely related: the 50% rule (check 2) is a single-point version of the full chi-squared QQ plot (check 3).
- Check 1 tests the **marginal** distributions; checks 2 and 3 test the **joint** multivariate structure.
- **Outliers** show up in check 3 as points deviating strongly from the line at the upper end of the plot.
- For formal testing (not just graphical), consider the **Mardia test** or **Henze–Zirkler test**.